

Ziad Elhussein

AI & Machine Learning · Data Science

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SUMMARY

AI/ML-focused Computer Science graduate (First Class Honours, University of London) working as an Associate Software Consultant in banking technology. I build machine-learning systems end to end: data collection, leak-free evaluation, model training and web deployment. My graduation project, Pawlytics, combines three deep-learning models (text, audio, image) in one product on a Node.js and Flask stack, with every result reported on held-out test data. 21 public projects with code and live demos on GitHub.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, C++, C#, Bash
Machine learning & AI	PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, OpenCV, NLP, computer vision, audio ML (YAMNet), genetic algorithms, model evaluation, feature engineering, class-imbalance handling
Data	pandas, NumPy, Matplotlib, Jupyter, data cleaning and statistics, MySQL, SQLite, MongoDB
Software & backend	Flask, Node.js, Express, REST APIs, HTML, CSS, JavaScript, Git and GitHub, Docker (basics)
Testing & performance	Apache JMeter, Playwright, performance testing, test automation, Temenos T24

EXPERIENCE

Associate Software Consultant · Qestit

Dec 2025 – Present

- Design and run Apache JMeter load-test scenarios for banking portals that simulate real user load and validate system behaviour against defined exit criteria.
- Analyse response times, error rates, throughput and stability indicators to pinpoint bottlenecks and support release sign-off.
- Write Playwright automation for Temenos T24 banking workflows, re-validating critical screens and business processes after every update, fix or release.

PROJECTS

Pawlytics: multimodal AI pet-health assistant · Graduation project

github.com/ziadelh/pawlytics

Stack: Python, PyTorch, TensorFlow, Hugging Face, Flask, Node.js, Express, MongoDB

- Built and integrated three deep-learning models behind an Express web app and a Flask service: Bio_ClinicalBERT text classifier (20 conditions, 63.8% held-out accuracy), YAMNet + MLP audio classifier (8 classes, 71.9%, macro-F1 0.71) and EfficientNet-B0 image classifier (9 classes, 73.2%). A rule-based fusion step merges them into one report with urgency, confidence and treatment suggestions.
- Owned the data pipeline: merged and de-duplicated about 24.8k images from 20+ public sources, split the 51,841-record text dataset by dog to prevent leakage, and sourced and cleaned 953 audio clips. Used class-weighted loss, label smoothing and augmentation to handle imbalance.
- Evaluated only on held-out test data and documented limits in model cards, including that a more realistic text run scored 42.2%.

Fake-news detection · Natural Language Processing

github.com/ziadelh/fake-news-detection

Stack: Python, scikit-learn, NLTK, gensim

- Compared Naive Bayes with TF-IDF (94% accuracy) against Word2Vec with logistic regression (97%) using a leakage-free train/test pipeline.

Traffic car detection, tracking and counting · Computer vision

github.com/ziadelh/traffic-car-tracking

Stack: Python, OpenCV

- Detect, track and count cars in traffic-camera footage with frame differencing, background subtraction and Kalman filters; counts match a manual count on both videos.

Evolutionary robotics

github.com/ziadelh/artificial-intelligence-mountain-climbing-creature

· Mountain-climbing creatures

Stack: Python, PyBullet, genetic algorithms

- Genetic algorithm that evolves virtual creatures to climb a mountain in a PyBullet physics simulation: tree-shaped genome, multi-term fitness rewarding vertical progress and stability, tournament selection, CSV logging and fitness charts.

Offline speech-recognition assistant · Audio / speech

github.com/ziadelh/speech-recognition-assistant

Stack: Python, Vosk, SciPy

- English, Spanish and Italian airport-phrase recognition with Vosk; 4.5% mean word error rate with the large models, plus a noise-filtering experiment and WER evaluation.

Machine-learning and data-science studies · Data science

github.com/ziadelh

Stack: Python, pandas, scikit-learn, TensorFlow/Keras

- Loan approval: decision tree (96.8% accuracy) vs KNN written from scratch, PCA and nested cross-validation. House prices: tuned Keras network with dropout and early stopping (R^2 0.89). Movie revenue: regression on 9,700 TMDb movies (linear R^2 0.68, polynomial 0.73).

Audio and media tools · Signal processing

github.com/ziadelh/web-audio-apps

Stack: JavaScript, p5.js, Meyda, Python, ffmpeg

- Browser audio effects processor, audio CAPTCHA and music visualisers (live demo); audio steganography; lossless audio compression with Rice coding; automatic film-format checking and conversion with ffprobe and ffmpeg.

EDUCATION

BSc (Hons) Computer Science, First Class Honours

Sep 2025

University of London (academic direction: Goldsmiths, University of London) · Graduated September 2025 · Specialisation: Artificial Intelligence and Machine Learning

Relevant coursework: Artificial Intelligence, Machine Learning, Neural Networks, Natural Language Processing, Intelligent Signal Processing, Data Science, Databases and Advanced Data Techniques, Algorithms and Data Structures, Object-Oriented Programming, Agile Software Projects, Web Development, Computer Security

ADDITIONAL EXPERIENCE

Ambassador · European Universities in Egypt

Jan 2022 – Apr 2024

Led student workshops, ran university events and supported digital content and outreach.

Supervisor · Family-Owned Contracting Company

Jun 2020 – Present

Coordinate on-site teams, timelines, suppliers, procurement and cost tracking.

Equestrian Coach · Armed Forces Equestrian Club

Jan 2023 – Nov 2024

Coached riders and prepared students for competitions.

Languages: Arabic · English · German (learning)